

Jakub Sokal

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Profile Summary

Software engineer with 12 months of commercial experience at AMCS Group, working on live payment systems, diagnosing a critical double-charge defect in production and building end-to-end Stripe and SEPA integrations. Named student researcher on the EU-funded Prompt Me project at Lero, University of Limerick, designing and building a multi-agent LLM orchestration framework for AI-driven Requirements Engineering (RE). Hands on experience building and optimising backend systems under real production constraints with focus on large scale infrastructure and distributed systems. Graduating August 2026.

Education

University of Limerick

Limerick, Co. Limerick

Bachelor of Science in Computer Science | Expected August 2026 | First Class Honours (QCA: 3.44/4.0)

Course Work: Database Systems, Advanced Programming Concepts and Practices, Data Structures & Algorithms, Software Architecture, Applied System Design, Operating Systems, Software Testing and Inspection, Event Driven Programming.

Professional Experience

AMCS Group

Limerick, Co. Limerick

Associate Software Developer

June 2025 – August 2025

- Designed and delivered production Stripe Card and SEPA payment integrations from technical design through to deployment.
- Diagnosed and fixed a critical double-charge defect in the live payment flow by tracking down the root cause, patching it and shipping to production.
- Optimised a transaction endpoint handling 5,000+ records, reducing response time by 40% from approximately 10 minutes to under 6 minutes.
- Identified and documented the root cause of a serialization bottleneck in the RESTful API layer, producing an analysis that pinpointed the exact failure point and directly enabled the fix.
- Built Azure Data Factory pipelines for ETL processes and managed CI/CD deployments across dev, test, and production Azure environments.

AMCS Group

Limerick, Co. Limerick

Intern Software Developer

May 2024 – January 2025

- Built out unit test coverage across core payment and API components in a large-scale C#/.NET and Angular codebase, writing 500+ tests and consistently maintaining the team's 80% per-PR coverage requirement.
- Worked on ACI Cards payment integration throughout a live refactoring cycle, shipped bug fixes, features, and tests while the code base was actively changing.
- Traced a data integrity bug in the Payment Express feature back to an upstream service misappending results, identifying the root cause of missing column data in the client-facing UI and documenting findings for resolution.
- Extended and maintained 30+ RESTful APIs across live payment and data flows, shipping to production across multiple sprint cycles.

Project Experience

Final Year Project (SimOrch)

University of Limerick

Orchestration Framework for Multi-Agent RE Simulation | [GitHub](#) | [Link](#) | Python | React | Typescript September 2025 – April 2026

- Named student researcher on the EU-funded Prompt Me project at Lero, University of Limerick, supervised by Dr. Jacek Dabrowski, investigating the feasibility of AI-driven Requirements Engineering simulation.
- Designed and built a multi-agent orchestration framework in Python coordinating GPT-4-powered Requirements Engineer, Stakeholder, and Analyst Agents with shared memory, persona-based configuration, and structured logging to produce reproducible and traceable RE simulation outputs.
- Evaluated the framework across seven dimensions, achieving 100% feasibility and 100% traceability across all ten simulation runs, with structural reproducibility of 100% and ground truth requirement coverage of 83.3%.
- Identified a limitation in the current simulation: LLM-based RE agents failed to elicit non-functional requirements under the tested persona configuration, highlighting an area for further investigation in dialogue-driven elicitation.
- Research findings contributed to an in-progress international collaboration spanning University of Limerick, Trinity College Dublin, and The Open University.

Software Architecture Project (Campus2Company)

University of Limerick

Microservices Platform | [GitHub](#) | Java | Spring Boot | React | Kafka | Docker

February 2026 – April 2026

- Led development across the team, owning code reviews and driving architecture decisions from initial design through to completion.
- Designed and built a Java Spring Boot microservices architecture with an API gateway to decouple services and manage inter-service communication.
- Implemented a full JWT authentication system including access tokens, refresh tokens, and long-lived tokens for persistent session management across services.
- Integrated Kafka for asynchronous event-driven communication between microservices, decoupling producers and consumers across the platform.
- Containerised core services using Docker to ensure consistent deployments across environments.
- Built a responsive React frontend supporting interactive workflows across multiple user roles such as students, universities, and businesses.

Skills

Languages: Java, Python, C#, JavaScript, TypeScript, C++.

Frontend: Next.js, React, Angular, HTML, CSS.

Backend: RESTful APIs, Spring Boot, .NET Framework, Entity Framework, Kafka, SQL, PostgreSQL, MongoDB, Supabase, JWT Authentication.

Testing: Postman, NUnit, JUnit, Jasmine, pytest

AI / LLM: LangChain, OpenAI API, Multi-Agent Orchestration.

Infrastructure & Tools: CI/CD, Docker, Azure, Git, Azure Data Factory.

Soft Skills: Problem Solving, Critical Thinking, Communication, Teamwork, Agile Methodologies, Attention to Detail, Adaptability.

References

Available upon request.